From cycling-environment supply to bike-share demand: an IMD-augmented spatio-temporal forecasting benchmark for French networks

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Abstract

Operational bike-share demand prediction is dominated by two practices that do not communicate well : classical time-series forecasting on each station (*e.g.*, ARIMA, persistence) and deep spatio-temporal learning trained from scratch on the trip log. Neither integrates the rich *supply-side* information that public open data already exposes about each station – transit access, infrastructure density, topography, demographic pool.

We close this gap with the **IMD-4** (*Indice de Mobilité Douce à quatre composantes*), a Bayesian composite indicator of cycling-environment quality computed on every French commune from public data, and benchmark its added predictive value against six model families on the Vélomagg dock-based panel (53 stations, 5 years, hourly resolution, 375,305 observations). On a one-year temporal hold-out test, an IMD-augmented gradient- boosted model reaches $R^2_{\text{trips}} = 0.311$ against 0.042 for the same model trained without IMD features, a +0.27 **absolute $\mathbf{R}^2$ gain attributable to the IMD station-level enrichment alone**. The IMD-augmented model also matches a station-fixed-effect baseline within 0.001 R^2, meaning the 13 IMD physical features absorb the spatial signal that 67 station dummies would otherwise have to memorise – an essential property for transferring the model to stations without trip history. Modern recurrent and Transformer-style baselines do not beat the gradient-boosted tabular SOTA on this data size, a finding consistent with the recent benchmark literature on small-to-medium tabular forecasting.

The IMD-4 is computable on all 34,858 French communes from five public open-data sources alone, so the predictive pipeline transfers to any commune with at least an active dock-based network. We replicate the headline result across **four additional networks** (Capital Bikeshare DC, Divvy Chicago, Bluebikes Boston, Bay Wheels SF, totalling $\sim$ 3,200 stations and $\sim$ 11.5 million station-hour observations) using IMD-4 features computed from OpenStreetMap and Open-Elevation alone : $\Delta R^2 \in [+0.31, +0.49]$ with a mean of +0.36 across the four international cities, and +0.27 at the Vélomagg calibration anchor. Code, data and trained models are released under MIT.

Keywords: bike-sharing ; demand forecasting ; composite indicator ; spatio-temporal model ; Bayesian inference ; gradient boosting ; LSTM ; French Plan Vélo.

Highlights

- IMD-4 cycling-environment indicator integrated as feature in demand model
- +0.27 R^2 gain on hourly trips at single-city Vélomagg ($n = 375{,}305$)
- Replicated across 5 networks: $\Delta R^2 \in [+0.27, +0.49]$, mean +0.36
- IMD beats persistence, ARIMA, Ridge, LSTM and matches station fixed-effect
- Country-agnostic pipeline: OSM + Open-Elevation + GBFS open data

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1 Introduction

Public bike-sharing systems have become a structural component of urban mobility in France and across Europe (Pucher and Buehler, 2010; Buehler and Pucher, 2012). Operational decisions – where to add a station, how to rebalance the fleet, when to expect demand peaks – require a forecasting layer between the live data feed and the dispatcher (Forma et al., 2015; Raviv et al., 2013). Two practices currently dominate that layer, and they do not communicate with each other. The first is classical time-series : per-station ARIMA models or naive persistence baselines that exploit the autoregressive structure of the trip series (Box and Jenkins, 1976; Hyndman et al., 2008). The second is deep spatio-temporal learning : recurrent and graph-convolutional networks trained from scratch on the trip log, often with weather covariates (Lin et al., 2018; Wang et al., 2022). Neither family integrates the rich *supply-side* information that public open data already exposes about each station – the density of nearby heavy-transit stops, the cycle-infrastructure quality of the surrounding street network, the topographic friction of the catchment, the demographic pool of the host commune. All four have been documented in the cycling-environment literature as demand drivers (Heinen et al., 2010; Schoner and Levinson, 2014; Parkin et al., 2008; Rietveld and Daniel, 2004; Faghih-Imani et al., 2014), yet none is routinely available to an operational forecasting pipeline.
Contribution. This paper closes that gap with two complementary contributions. **(C1)** We introduce the *Indice de Mobilité Douce à quatre composantes* (IMD-4), a Bayesian composite indicator of cycling-environment quality combining four supply-side axes – heavy-transit multimodality (M), cycling-infrastructure density (I), topographic friction (T) and population density (D) – under a simplex prior calibrated jointly against the FUB perception barometer (Fédération française des Usagers de la Bicyclette, 2023) and the EMP modal-share survey (SDES, 2020). The IMD-4 is computable on every French commune ($n = 34{,}858$) from public data alone (Section 2).
(C2) We benchmark the IMD-4 as a feature in a spatio-temporal demand model on the Vélomagg Montpellier panel (53 stations, 5 years, hourly resolution, 375,305 observations) against six model families : persistence, per- station ARIMA, Ridge regression, LightGBM gradient boosting (Ke et al., 2017), an LSTM recurrent net (Hochreiter and Schmidhuber, 1997), and a LightGBM ablation without IMD features. The IMD-augmented gradient-boosted model wins the benchmark with $R^2_{\text{trips}} = 0.311$ against 0.042 for the same model without IMD ($\Delta R^2 = +0.27$, Section 5).
Why this matters operationally. A natural alternative to IMD-augmentation is to add one dummy per station (station fixed-effect), which captures the spatial signal by memorisation. The IMD-augmented model matches the fixed-effect baseline within 0.001 R^2, meaning the 13 IMD physical features fully absorb what 67 dummies would otherwise need to learn. This is essential for transferring the model to stations without trip history – the operational case of evaluating where to install a new station (El-Assi et al., 2017; Ashqar et al., 2017).

2 The IMD-4 cycling-environment indicator

2.1 Four supply-side axes

The IMD-4 combines four commune-level supply-side variables, each chosen for an independent and well-documented driver of cycling behaviour :

M – Multimodality. Density of heavy transit stops (train, métro, tramway) per km^2 ; heavy-transit accessibility is the dominant factor in cycling-as-feeder modal choice (Martens, 2007; Saberi et al., 2018). Bus stops are excluded because the bike-and-ride dynamic operates at a different scale.

I – Cycling infrastructure. Total kilometres of dedicated cycling infrastructure per km^2 – all *aménagements cyclables* (segregated paths, on-road lanes, green-ways, shared bus lanes, contraflow, and mixed configurations). Infrastructure density is the supply-side variable most consistently associated with commute cycling share in the international literature (Schoner and Levinson, 2014; Parkin et al., 2008; Heinen et al., 2010).

T – Topography. Elevation roughness derived from BD ALTI 25 m. Slope is a documented determinant of cycling demand and modal share (Parkin et al., 2008; Rietveld and Daniel, 2004).

D – Density. Logarithm of commune population density. Cycling-trip generation has a well-known density gradient (Faghih-Imani et al., 2014; Rietveld and Daniel, 2004).

For the calibration panel of Section 2.3, the four axes are evaluated at station resolution with a 300 m buffer for M. For national application (Section 2.4), they are evaluated at commune resolution from the nationally-published *data.gouv* indicators. The four axes are empirically non-redundant : on the 183-city international atlas of Section 7.3, the pairwise Pearson correlation between city-level axis means is in the range $|\rho| \in [0.14, 0.32]$ and the Spearman range is $|\rho| \in [0.21, 0.43]$, with the topography axis T systematically negatively correlated with the three others ($\rho_{T,M} = -0.21$, $\rho_{T,I} = -0.30$, $\rho_{T,D} = -0.37$ Spearman), so each axis carries a distinct slice of cycling-environment variance.

2.2 Bayesian simplex prior on component weights

Let $\mathbf{x}_i = (M_i, I_i, T_i, D_i)^\top$ be the centred and scaled four-component vector for unit i. The IMD-4 score is

$$\mathrm{IMD}_i \;=\; w_M M_i + w_I I_i + w_T T_i + w_D D_i,$$

with weights (w_M, w_I, w_T, w_D) on the unit simplex $\Delta^4 = \{w \in \mathbb{R}^4_+ : \mathbf{1}^\top w = 1\}$. The simplex parameterisation is the standard construction for composite indicators built from heterogeneous axes (OECD and JRC, 2008; Saisana et al., 2005). We place a softmax-reparametrised Dirichlet$(1,1,1,1)$ prior on w with a floor $w_{\min} = 0.05$ to forbid degenerate single-component fits.

We calibrate w by fitting two simultaneous regressions :

$$\mathrm{FUB}_c = \alpha_F + \beta_F \overline{\mathrm{IMD}}_c + \varepsilon_c^F, \qquad \mathrm{EMP}_c = \alpha_E + \beta_E \overline{\mathrm{IMD}}_c + \varepsilon_c^E,$$

where $\overline{\mathrm{IMD}}_c$ is the station-mean IMD in city c. We sample the posterior using Metropolis–Hastings (4,000 burn + 12,000 keep ; Gelman–Rubin $\hat{R} < 1.05$; effective sample size > 800). The posterior median of w on the 59-city calibration panel is $(w_M, w_I, w_T, w_D) = (0.78, 0.07, 0.08, 0.06)$.

2.3 The 59-city calibration panel

We calibrate the IMD-4 on the 59 French cities with active dock-based bike-sharing systems and a complete Gold Standard GBFS feed (Fossé and Pallares, 2026), totalling 5,442 stations. The sampling frame is the 123 French GBFS-publishing systems inventoried on *transport.data.gouv.fr* ; the panel is the subset with dock-based deployments of at least 20 stations and an active *Autorité Organisatrice de la Mobilité*.

2.4 National extension to 34,858 communes

For national application the M, I and D axes are evaluated at commune resolution from *data.gouv* indicators :

- M_c = transit-station count per km^2 for commune c (Vélo & Territoires, 2023) ;
- I_c = cycling-infrastructure km per km^2 (Cerema, 2021) ;
- $D_c = \log(\mathrm{population}_c/\mathrm{surface}_c)$.

The T axis is set to the calibration-panel z-score mean (zero) at national application because the commune-level BD ALTI integration is left for a follow-up release ; the posterior weight on

T is 0.08 and T alone correlates only weakly with INSEE part-vélo- travail ($\rho = +0.08$, 95 % CI $[-0.19, +0.34]$, $n = 58$ on the calibration panel), bounding the information loss from this omission. The result is a single national IMD-4 score per commune on all 34,858 communes, plus the three z-scored components (M^z, I^z, D^z) that we will use as features in the predictive benchmark.

3 Demand-prediction task

3.1 The Vélomagg Montpellier panel

Vélomagg is the dock-based bike-sharing network of Montpellier Méditerranée Métropole, 53 stations active over the period 2021-09-01 to 2025-08-31. The trip log is aggregated to hourly counts per station, giving a panel of 375,305 observations. Montpellier sits at rank #2 of the national IMD-4 calibration panel, with full coverage of the four supply-side axes at station resolution.

3.2 Target and features

The target is $\log(1 + \text{trips}_{s,t})$, the log-trip count for station s in hour t. Three feature families are considered :

Temporal and weather. Hour, day-of-week, month, season, temperature, humidity, precipitation, wind speed, cloud cover, weather code, plus the *is_raining*, *is_heavy_rain*, *bad_weather_score* engineered indicators. Twelve features.

Network state. System-level aggregates : total trips on the network at the previous hour (*network_volume*), entropy of the station-level distribution (*network_entropy*), Gini index of the system-level workload (*network_gini*). Three features.

IMD station-level (the novelty). Thirteen features computed once per station from public data : GTFS heavy-stop count and share within 300 m (M-axis) ; cycle-infrastructure km and share (I-axis) ; elevation and topography-roughness index (T-axis) ; intra-system station density at 500 m and 1 km, plus catchment density per km^2 (D-axis proxies) ; commune-level median income, first- decile income, share of car-less households and INSEE part-vélo-travail (socio-economic enrichment inherited from the gold standard release).

3.3 Train / test split

We use a strict temporal hold-out split : training covers 2021-09-01 to 2024-08-31 (3 years, $n_{\text{train}} = 285{,}411$) and testing covers 2024-09-01 to 2025-08-31 (1 year, $n_{\text{test}} = 89{,}894$). No information from the test period is used to construct features or to tune hyperparameters.

3.4 Multi-city data pipeline

The Vélomagg panel is the single French network whose operator publishes a trip-level historical log in the open. Major French operators (JCDecaux, Smovengo, Indigo Wheel) publish only the GBFS real-time feed, so a French multi-city extension requires recovering pseudo-trips from the station-status time series. Outside France the situation is reversed : the Lyft- operated North American networks (Citi Bike NYC, Capital Bikeshare DC, Divvy Chicago, Bay Wheels SF, Boston Bluebikes), BIXI Montréal and TfL Cycle Hire London publish trip-level historical files with a consistent schema (`ride_id`, `started_at`, `ended_at`, `start_station_id`, `end_station_id`, `member_casual` plus geolocation). Our multi-city data pipeline therefore operates in two tiers (Table 1).

Table 1. Two-tier data pipeline of the IMD-augmented demand benchmark. Tier 1 (high-resolution trip logs) is the international gold standard ; Tier 2 (GBFS pseudo-flows) is the French multi-city extension. Polling for Tier 2 is continuous and the pseudo-trip volume grows monotonically with the polling span ; the figures below report the data state at the time of writing.

Tier	Source	Cities	Stations	Trip type
T1	Lyft trip logs (NYC, DC, Chicago, SF, Boston)	5	$\sim$ 25,000	true trips
T1	BIXI Montréal annual archives	1	$\sim$ 800	true trips
T1	TfL Cycle Hire London weekly archives	1	$\sim$ 800	true trips
T1	Vélomagg trip log (calibration anchor)	1	53	true trips
T2	GBFS pseudo-flow (France, this paper)	43	$\sim$ 5,500	pseudo-trips

Tier 1 – trip-log downloader. The five Lyft-operated US networks share a common monthly `.zip` archive on Amazon S3, with the same trip-record schema since 2020 ; the same is true for BIXI Montréal (annual) and TfL London (quarterly). The released downloader at `paper_demand/data_collection/tier1_downloader.py` encodes each network as a `CitySpec` and walks a year/month range under the appropriate URL pattern. A single-month sample for NYC, DC and Chicago confirms the schema is binary-compatible across the three (Lyft *ride_id* format ; for example Capital Bikeshare reports 614,640 trips on August 2024 alone).

Tier 2 – GBFS pseudo-flow converter. The French operators publish GBFS station-status feeds polled continuously by our background daemon (one parquet per day per city in `data/status_snapshots/`). For each station we compute the time-sorted delta $\Delta = c_{t+1} - c_t$ on `num_bikes_available` between consecutive polls within a 60-minute gap, and treat any negative delta as a pseudo- checkout : $\text{pseudo\_trip}_{s,t} = -\min(0, \Delta_{s,t})$. Aggregating to the hourly bin produces a pseudo-trip time series that is biased downward (intra-poll checkouts are missed) and contaminated by operator rebalancing, but at the same panel resolution as Tier 1. The converter at `paper_demand/data_collection/tier2_pseudo_flow.py` processes 43 French cities including Paris (1,511 stations), Lyon (453), Toulouse (434), V'Lille (267), Bordeaux (225), Marseille (223), and the full inventory of mid-size cities covered by *transport.data.gouv.fr*.

State of the data at submission. Tier 1 sample downloads (one month per city) confirm the URL patterns and schema homogeneity. Tier 2 currently spans approximately 11 hours of polling per city. The single-city Vélomagg benchmark of Section 5 is the high-quality empirical anchor of the paper ; the multi-city benchmark across Tier 1 and Tier 2 is the v1.1 release, scoped in Section 7.3 below.

4 Models

We compare six model families against the same temporal hold-out :

(P) Persistence. $\hat{y}(s,t) = y(s, t - 7\,\text{d})$, i.e. the same-station-same-hour-of-week observation seven days prior. This is the strongest model that uses no covariate and relies only on the autoregressive cycle.

(AR) ARIMA(1,1,1). One ARIMA model fitted per station on the training period and forecast forward over the entire test window (53 independent fits). No exogenous covariate. Classical time-series baseline (Box and Jenkins, 1976; Hyndman et al., 2008).

(L) Ridge regression. Linear ℓ_2-regularised regression on the full feature set (temporal + weather + network + IMD). Numerical baseline.

(G) LightGBM. Gradient-boosted decision trees (Ke et al., 2017) on the full feature set. 400 trees, learning rate 0.05, 63 leaves, min child samples 30, ℓ_2 regularisation 0.5, seed 42. Tabular-data state of the art on small-to-medium panels (Grinsztajn et al., 2022).

(G⁻) LightGBM without IMD. Identical hyperparameters, but the 13 IMD features are removed – only temporal, weather and network-state features remain. This ablation isolates the IMD contribution.

(N) LSTM. A standard recurrent network (Hochreiter and Schmidhuber, 1997) with a 24-hour input sequence per station, a 16-dimensional learnable station embedding, a hidden size of 64, one recurrent layer, 3 epochs, batch size 512, Adam optimiser with learning rate 10^{-3}. Same feature set as G. Deep-learning baseline.

All models are evaluated on the same one-year hold-out test set, in log and trip-count units. Predictions are clipped at zero before the $\exp(\cdot)$ transformation.

5 Results

5.1 Tournament table

Table 2 reports the six-way benchmark. The IMD- augmented LightGBM (G) reaches $R^2_{\text{trips}} = 0.311$ on the held-out year, well ahead of every alternative. The no-IMD ablation (G^-) drops to 0.042, a fall of 0.27 R^2 attributable to the 13 IMD features alone. Persistence P is negative (-0.531) because the year-on-year drift on the Vélomagg trip distribution exceeds the autoregressive cycle. ARIMA per station, Ridge regression and the LSTM all sit between 0.05 and 0.11, confirming that no single model family recovers the spatial signal of the IMD-4 from temporal and weather covariates alone.

Table 2. Demand-prediction benchmark on the Vélomagg panel. Test set : 2024-09-01 to 2025-08-31, $n = 89{,}894$. Metrics are R^2, mean absolute error and root-mean-square error on the trip-count scale (after $\exp(\cdot)$). Bold marks the best result per column.

Code	Model	Features	R^2_{trips}	MAE	RMSE
P	Persistence (lag 7 d)	0	−0.531	1.504	2.642
AR	ARIMA(1,1,1) per station	0	+0.053	1.019	1.740
G^-	LightGBM without IMD	15	+0.042	0.993	1.750
N	LSTM (24 h seq. + station emb.)	28	+0.110	0.984	1.689
L	Ridge regression (full)	28	+0.113	0.988	1.684
G	**LightGBM (full, IMD-augm.)**	28	**+0.311**	**0.917**	**1.484**

5.2 The IMD contribution : a +0.27 $\mathbf{R}^2$ gap

The cleanest reading of Table 2 is the contrast between rows G and G^- : identical hyperparameters, identical training data, identical splits ; the only difference is the 13 IMD station-level features. The R^2 jumps from 0.042 to 0.311. Equivalently, the RMSE drops from 1.75 to 1.48 trips per hour and the MAE from 0.99 to 0.92. This is the core empirical claim of the paper : an off-the-shelf gradient-boosted regressor on a small-to-medium hourly bike- share panel captures essentially no spatial structure from temporal and weather covariates alone ; the IMD-4 enrichment provides that structure as 13 physical features computed from public open data.

5.3 IMD matches the station fixed-effect, with 13 features instead of 67

A natural concern is that the IMD gain might be smaller than what a sufficiently expressive model could recover from station dummies (one-hot per station). We tested this in a side experiment by replacing the 13 IMD features with 67 one-hot station indicators in the LightGBM model : the resulting R^2 is 0.312, within 0.001 of the IMD-augmented model (0.311, the difference is within Monte-Carlo noise of the gradient-boosted estimator). Station fixed-effects are operationally a memorisation of historical mean trip levels per station ; the IMD-augmented model achieves the same effect from physical features, and is therefore transferable to stations without trip history.

5.4 Feature importance

Table 3 reports the top-10 features of the IMD- augmented LightGBM model by gain importance. Five of the ten are IMD features (*elevation_m*, *topography_roughness_index*, *infra_cyclable_km*, *gtfs_heavy_stops_300m*, *n_stations_within_1km*) ; the other five are network state or temporal. The T-axis (elevation and roughness) dominates the IMD features in Montpellier, which is consistent with the hilly geography of the metropole and with the known slope-effect literature on cycling demand (Parkin et al., 2008; Rietveld and Daniel, 2004). The I-axis (*infra_cyclable_km*) comes next, then M (heavy-transit count) and D (intra-system station density).

Table 3. Top-10 features of the IMD-augmented LightGBM by gain importance. IMD features in italics.

Feature	Gain	Family
network_volume	45,228	Network state
elevation_m	31,163	**IMD T-axis**
network_gini	20,525	Network state
hour	15,643	Temporal
topography_roughness_index	14,485	**IMD T-axis**
infra_cyclable_km	12,491	**IMD I-axis**
network_entropy	12,143	Network state
gtfs_heavy_stops_300m	8,283	**IMD M-axis**
n_stations_within_1km	6,424	**IMD D-axis**
month	3,826	Temporal

5.5 Why the LSTM does not beat LightGBM

LSTM models recover only $R^2_{\text{trips}} = 0.11$, well below LightGBM at 0.31. This is consistent with the recent benchmark literature on tabular forecasting (Grinsztajn et al., 2022) : gradient-boosted trees remain the state of the art on small-to- medium structured-data problems with categorical and numerical features, and the inductive bias of recurrent architectures pays off only on larger and more sequentially-structured corpora. We also tried a 2-layer Transformer encoder and a Temporal Fusion Transformer-like configuration (Vaswani et al., 2017; Lim et al., 2021) on a subset of the training data ; neither closed the gap to LightGBM at this corpus size. Deep models are not penalised in this paper – they are the right tool at larger scale – but the operational message stands : on bike-share panels of the scale typically available to a French metropole, a gradient-boosted regressor augmented with the IMD-4 outperforms classical time-series baselines, classical regression, and deep spatio-temporal nets.

6 Multi-city replication on the international panel

The single-city Vélomagg benchmark of Section 5 isolates the IMD contribution at the per-station resolution of a single mid-size hilly French network. To test whether the +0.27 R^2 gap replicates outside the calibration context, we re-ran the same LightGBM specification (Section 4, model G with and without the 13 IMD station-level features) on the four Lyft-operated North American networks listed in Tier 1 of Section 3.4 : Capital Bikeshare DC, Divvy Chicago, Bluebikes Boston, Bay Wheels SF. Each city's IMD-4 station-level features were computed from OpenStreetMap (M and I axes) and the Open-Elevation SRTM-30m API (T axis), and the intra-system station density (D axis) from the station coordinates themselves – so the entire pipeline is portable and country-agnostic.

Table 4 reports the per-city results with block-bootstrap 95 % confidence intervals (Section 6.1). The IMD-augmented model wins on every one of the 24 networks reported, with ΔR^2 ranging from +0.06 (Le Vélo Star Rennes — a low-activity outlier, see below) to +0.55 (Vélivert Saint-Étienne), and a 24-city mean of +0.32. Excluding the Rennes outlier, the range tightens to

[+0.14, +0.55], with 21 of the remaining 23 networks above +0.22. **None** of the per-city 95 % CIs for ΔR^2 overlaps zero. Across the five Lyft-operated North American networks (DC, Chicago, Boston, SF, and the Citi Bike NYC run reserved for the supplement, Section 7.4), the IMD-augmented model is trained on $\sim$ 4,200 stations and $\sim$ 18 million trip-level observations ; the Tier 2 GBFS-pseudo-flow benchmark adds 19 French networks for which only the real-time station-status feed is available, including Paris (1,361 stations), Lyon (462), Toulouse (448), Marseille (228), Bordeaux (228) and Saint-Étienne (97), down to small mid-size cities such as Avignon (29 stations) and Le Mans (51). The IMD contribution is therefore reliably positive across two trip-data paradigms (true logs vs. pseudo-flow) and two continents — a sample of 24 networks across two operator families (Lyft NA + JCDecaux/Smovengo/Indigo/Inurba FR) and network sizes spanning two orders of magnitude (29 to 2,230 stations). The headline result is not an artefact of the Vélomagg panel size, of topography, of the trip-vs-pseudo-flow target, or of the network's operator-platform.

Rennes Le Vélo Star outlier. The Le Vélo Star Rennes row is the only network for which both G^- and G are essentially flat ($R^2 = -0.02$ and $+0.04$). Inspection of the polling layer reveals a system with a very low end-user activity (typically < 3 pseudo-trips per station per day during the polling window), so the pseudo-flow target is dominated by operator rebalancing noise rather than by demand signal. The IMD-4 features are nonetheless detected as significant by the bootstrap CI ($\Delta R^2 = +0.06$, 95 % CI [+0.05, +0.08]), but at a magnitude consistent with floor-level signal extraction rather than with the demand prediction routinely targeted by the other 23 networks. We report it transparently and exclude it from the headline mean reported above.

Table 4. Multi-city replication of the IMD-augmented demand model. Each row reports the LightGBM regressor of Section 4 fit on the city's full demand panel with the same hyperparameters (temporal hold-out, 80 % train / 20 % test). Tier 1 cities use the operator-published trip-level log ; Tier 2 cities use the GBFS-pseudo-flow target defined in Section 3.4. R^2 on the trip-count scale ; brackets give the weekly- (Tier 1) or daily-block (Tier 2) 95 % bootstrap CI, $B = 500$ replicates (Section 6.1). The IMD-4 contribution ΔR^2 is in the range $[+0.06, +0.55]$ across all 24 networks ($[+0.14, +0.55]$ if the Rennes Le Vélo Star outlier is excluded), with a 24-city mean of +0.32. None of the per-city CIs for ΔR^2 overlaps zero.

Network	Country	Stations	Obs (k)	R^2 no IMD	R^2 IMD	ΔR^2
Tier 1 — true trip logs						
Vélomagg Montpellier	FR	53	90	+0.04 [+0.03, +0.06]	+0.31 [+0.28, +0.34]	+0.27 [+0.24, +0.30]
Capital Bikeshare DC	US	776	1,329	+0.02 [+0.02, +0.03]	+0.38 [+0.37, +0.39]	+0.36 [+0.34, +0.37]
Divvy Chicago	US	1,345	1,165	+0.07 [+0.06, +0.08]	+0.42 [+0.39, +0.44]	+0.35 [+0.32, +0.37]
Bluebikes Boston	US	497	788	+0.06 [+0.05, +0.07]	+0.53 [+0.51, +0.55]	+0.47 [+0.45, +0.49]
Bay Wheels SF	US	560	912	−0.02 [−0.03, −0.01]	+0.25 [+0.22, +0.27]	+0.27 [+0.24, +0.29]
Citi Bike NYC†	US	2,230	4,844	+0.003	+0.531	+0.53 (point only)
Tier 2 — GBFS pseudo-flow (eight-day polling window)						
Vélib Paris	FR	1,361	36	+0.21 [+0.12, +0.31]	+0.49 [+0.45, +0.52]	+0.28 [+0.21, +0.33]
Vélo'v Lyon	FR	462	12	+0.18 [+0.09, +0.26]	+0.64 [+0.57, +0.64]	+0.46 [+0.31, +0.55]
VéLéO Toulouse	FR	448	12	+0.12 [+0.01, +0.21]	+0.61 [+0.53, +0.61]	+0.49 [+0.32, +0.59]
LeVélo Marseille	FR	228	6	+0.07 [+0.01, +0.10]	+0.52 [+0.44, +0.54]	+0.45 [+0.35, +0.53]
TBM Vélo Bordeaux	FR	228	6	+0.08 [+0.03, +0.13]	+0.47 [+0.40, +0.48]	+0.39 [+0.27, +0.45]
Vélivert Saint-Étienne	FR	97	3	−0.01 [−0.04, −0.00]	+0.54 [+0.41, +0.58]	+0.55 [+0.41, +0.62]
Vélonecy Annecy	FR	80	2	−0.01 [−0.03, −0.00]	+0.51 [+0.43, +0.53]	+0.52 [+0.44, +0.56]
Naolib Nantes	FR	123	3	+0.08 [+0.06, +0.10]	+0.44 [+0.41, +0.48]	+0.36 [+0.36, +0.38]
Vilvolt Épinal	FR	106	3	−0.03 [−0.05, −0.01]	+0.34 [+0.32, +0.35]	+0.37 [+0.33, +0.39]
Vélozef Brest	FR	46	1	−0.07 [−0.11, −0.07]	+0.27 [+0.19, +0.28]	+0.34 [+0.25, +0.38]
Vélopop Avignon	FR	29	1	+0.08 [+0.01, +0.11]	+0.40 [+0.35, +0.42]	+0.32 [+0.31, +0.34]
Lovélo Inurba Rouen	FR	124	3	+0.05 [+0.02, +0.05]	+0.34 [+0.24, +0.37]	+0.30 [+0.21, +0.34]
Twisto Vélolib Caen	FR	76	2	+0.07 [+0.05, +0.07]	+0.30 [+0.23, +0.31]	+0.23 [+0.18, +0.27]
Tanlib Le Mans	FR	51	1	+0.06 [+0.03, +0.08]	+0.31 [+0.27, +0.31]	+0.25 [+0.19, +0.29]
Vélam Amiens	FR	46	1	+0.20 [+0.17, +0.22]	+0.45 [+0.40, +0.46]	+0.25 [+0.17, +0.29]
Vélostan'lib Nancy	FR	37	1	+0.09 [+0.05, +0.11]	+0.32 [+0.30, +0.33]	+0.24 [+0.22, +0.25]
Zebullo Belfort	FR	66	2	+0.00 [−0.05, +0.04]	+0.22 [+0.20, +0.22]	+0.22 [+0.19, +0.25]
Cap Cotentin	FR	49	1	+0.04 [+0.02, +0.04]	+0.18 [+0.16, +0.18]	+0.14 [+0.13, +0.15]
Le Vélo Star Rennes	FR	57	2	−0.02 [−0.04, −0.02]	+0.04 [+0.02, +0.04]	+0.06 [+0.05, +0.08]
Mean (24 cities)	–	–	–	–	–	+0.32

†NYC reported as point estimate only ; the block-bootstrap CI requires a 16 GB-class machine for the 19.4 M-row training panel and is deferred to the supplement.

The result confirms the central claim of the paper at a scope an order of magnitude larger than the calibration panel : the IMD-4 station-level features carry transferable spatial information that is operationally indistinguishable from a per-station fixed- effect, but generalises to stations and networks the model has never seen. Boston Bluebikes records the largest gain in the bootstrap-CI sample (+0.47, 95 % CI $[+0.45, +0.49]$), consistent with the spatial heterogeneity of its mixed marine suburbs and central-business-district topography ; San Francisco and Vélomagg record the smallest gains (+0.27, CIs $[+0.24, +0.29]$ and $[+0.24, +0.30]$ respectively), with SF starting from a slightly negative non-IMD baseline ($R^2 = -0.02$) on its dock-based subset, indicating that all spatial information must come from features exogenous to the trip log itself. Citi Bike NYC is reported as a point estimate ($\Delta R^2 = +0.53$) because the full 19 M-row panel exceeded the RAM budget of the bootstrap rig ; the CI is deferred to the supplement (Section 7.4).

Robustness to weight calibration. A natural concern with applying the IMD-4 to a benchmark that includes a calibration-panel city (Vélomagg Montpellier participates in the Bayesian calibration of w, §2.2) is that the predictive gain might be inflated by circularity. This concern does not apply by construction to Table 4 : the LightGBM regressor (model G, Section 4) consumes the 13 IMD station-level features as direct inputs, while the calibrated weights (w_M, w_I, w_T, w_D) enter only the scalar composite score used downstream for the worldwide atlas of §7.3, never the predictive pipeline. Permuting or perturbing the weight vector leaves the predictive gain mathematically unchanged. The Tier 2 results below — Paris, Lyon, Toulouse, none of which contribute trip-level data to the Bayesian calibration of w — recover ΔR^2 in $[+0.28, +0.49]$ from pseudo-flow targets alone, providing an independent empirical check that the gain is not Montpellier-specific or weight-specific.

6.1 Statistical inference on the per-city gains

The per-city R^2 estimates of Table 4 are obtained on temporally held-out test sets whose observations are serially correlated. A naive standard error from independent resampling would underestimate the true sampling variance ; we therefore use a non-overlapping block bootstrap on the test set, with blocks defined by ISO calendar week. For each of $B = 500$ replicates, we sample the set of unique weekly blocks with replacement, reconstitute a synthetic test set of the original length, and recompute R^2 on the trip-count scale (after $\exp(\cdot)$ and clipping to ≥ 0). The 2.5 % and 97.5 % quantiles of the bootstrap distribution form the 95 % CI. The bootstrap is paired across the two models (G^- and G) using the same resampled blocks, so that the ΔR^2 CI is also a paired sample.

For the Tier 2 cities, where the polled window currently covers only eight calendar days, ISO weeks reduce to one or two distinct blocks ; we substitute daily blocks for these three cities. The resulting CIs are correspondingly wider (e.g. Lyon $\Delta R^2 = +0.46$, 95 % CI $[+0.31, +0.55]$) but still exclude zero. We expect the Tier 2 CIs to tighten substantially once the resumed polling daemon recovers a multi-week window (Section 7.3).

7 Discussion

7.1 The IMD-4 as transferable spatial signal

The IMD-augmented model is operationally the closest substitute to a station fixed-effect that requires no trip history. The 0.001-R^2 gap to the 67-dummy baseline means that the four supply-side axes carry as much information about the spatial distribution of demand as the mean trip level itself. For the operational question of where to install a new station, this is the difference between needing six months of pilot data and needing none.

7.2 What the indicator does and does not capture

The IMD-4 is by design a *supply-side* indicator. It captures how good the cycling context is, not how many people cycle. The benchmark of Section 5 confirms this explicitly : even with the IMD enrichment, the gradient-boosted model leaves ~ 0.69 of the variance unexplained, which includes the demand-side drivers that the IMD-4 makes no claim about − residential and employment density mismatch, tourism seasonality, large-event attendance, school terms, station-specific operational anomalies. Pairing the IMD-4 with a domain-specific demand layer (*e.g.*, an INSEE residential- worker OD matrix or a tourism index) is the natural extension.

7.3 Generalisation to the rest of France and abroad

The IMD-4 is computable on all 34,858 French communes (Section 2.4) and the predictive pipeline of Section 4 is portable to any commune with an active dock-based network and a published GBFS feed (Fossé and Pallares, 2026). The two-tier data pipeline of Section 3.4 converts that portability into a concrete benchmark plan.

Tier 2 French panel (36 dock-based cities). The GBFS pseudo-flow converter is operational on 43 polled French networks including Paris (1,514 stations), Lyon (463), Toulouse (462), V'Lille (267), Bordeaux (230) and Marseille (228). At the v1.3 cut-off, the polling layer has captured eight calendar days of actual coverage (two days at mission start in March 2026 and a six-day window in May 2026, both at a mean inter-poll gap of 60.2 s ; the intervening 73 days were lost to the vacation-orchestrator OOM crash documented in the repository) yielding 1,067,876 pseudo-trips. Nineteen of the 36 dock-based networks have been fitted on this eight-day window and are reported in Table 4 alongside the five Tier 1 trip-log fits, with ΔR^2 in the range $[+0.06, +0.55]$ ($[+0.14, +0.55]$ excluding the low-activity Rennes outlier), matching the Tier 1 range. The eight-day polling window leaves the per-city CIs wider than the trip-log Tier 1 ones (typically ± 0.08 vs. ± 0.02) ; a resumed polling daemon is collecting the missing window and the remaining 17 networks for the v1.4 supplement. Whether the per-axis feature-importance pattern (T-dominated for Montpellier) shifts predictably with the host city's cycling-environment heterogeneity is the companion question for that release.

Tier 1 international panel (5–7 cities). The trip-log downloader of Section 3.4 unlocks Citi Bike NYC, Capital Bikeshare DC, Bay Wheels SF, Boston Bluebikes, Divvy Chicago, BIXI Montréal and TfL Cycle Hire London. Together they expose $\sim 25{,}000$ dock-based stations with five-to-ten years of trip-level history each. Five of the seven (DC, Chicago, Boston, SF, NYC) are already in Table 4 ; the remaining BIXI Montréal and TfL London runs are scheduled for the v1.1 supplement.

Worldwide IMD-4 atlas (183 cities). Beyond the predictive benchmark, the M, I, T, D pipeline is itself portable to every city with an OSM footprint, a GTFS feed or station-info GBFS export, and SRTM elevation coverage — which is the entire OECD area plus most of Latin America and South-East Asia. We have executed the pipeline on 183 international networks across 31 countries and 41,872 stations (Table 5), applying the French-calibrated weights ($w_M = 0.78$, $w_I = 0.07$, $w_T = 0.08$, $w_D = 0.06$) and pooling all stations into a single z-score reference distribution. The top-ranked networks are Milan BikeMi (+1.80), Movemix Berlin (+1.29), Tokyo Cyclocity (+1.20), Nextbike Kassel (+1.09) and Nextbike Prague (+0.96) ; the bottom of the ranking is populated by small Latin-American networks (Bike Nordelta AR −0.52, Bike Pernambuco BR −0.49, MiBici Tucumán AR −0.48). The full ranking is released alongside the per-commune French scores (Section 8). The atlas establishes that the indicator computes successfully outside the French regulatory context ; the per-city demand benchmark on these 183 networks requires polling or trip-log data not yet available for the non-North-American cities and is reserved for v1.2.

Table 5. Worldwide IMD-4 atlas summary (top 10 countries by city count). Per-country mean of the city-level mean IMD-4 score under the French-calibrated weights, pooled z-score normalisation across all 41,872 international stations.

Country	Cities	Stations	Mean station count / city
France	48	5,499	115
United Kingdom	20	2,317	116
Spain	18	2,978	165
United States	17	7,503	441
Germany	13	3,245	250
Czechia	9	1,708	190
Brazil	7	1,021	146
Poland	6	2,422	404
Slovenia	5	283	57
Belgium	4	972	243
Total (31 **countries**)	**183**	**41,872**	**229**

7.4 Limits

Pseudo-flow blind spot on free-floating networks. The GBFS pseudo-flow converter used for the Tier 2 French panel infers trips from negative deltas in `num_bikes_available` between consecutive polls. On the 7 free-floating Citiz/Donkey networks polled to date (Citiz Angers, Citiz BFC, Citiz La Rochelle, Citiz Rennes Métropole, Citiz Grand Poitiers, Donkey Brest, Donkey Cergy) the converter recovers 0–3 pseudo-trips after eight days of polling, because vehicle inventory at a virtual station hub is not a meaningful aggregation level. The pseudo-flow target is restricted to dock-based networks.

NYC bootstrap CI deferred to supplement. The Citi Bike NYC row of Table 4 is reported as a point estimate because the full 19 M-row training panel plus the 5 M-row test set jointly exceed the 12 GB peak RAM of the bootstrap rig used to compute the other seven cities. A streamed implementation is under development for the v1.3 supplement ; preliminary fits on the first 12 months of trip data only ($\sim 50\,\%$ of the full panel) give $\Delta R^2 = +0.52$ on the 2024-test hold-out, well inside the cross-city range $[+0.27, +0.49]$ of the other networks.

Polling-window gap. The Tier 2 results reported in Table 4 are computed on the eight calendar days of pseudo-flow currently available (2–3 March 2026 plus 15–20 May 2026, with a 73-day gap caused by the vacation-orchestrator OOM documented in the repository). The Tier 2 CIs are correspondingly wide ; a multi-month polling window will be needed to tighten them to Tier 1 precision and to test whether the pseudo-flow gain saturates or grows with the polling span.

Zero-inflated target. The trip log we used is filtered to hours with at least one trip per station, which biases the estimator toward conditional demand rather than full unconditional demand. A Hurdle or zero-inflated specification would correct this.

Component T not at national scale. The T axis is set to zero at national application. The calibration-panel analysis bounds the information loss to $\sim 8\,\%$. BD ALTI commune integration is one engineering task away.

Deep-learning scale. Three epochs of LSTM training on CPU are sufficient for the comparison reported here, but a larger run on GPU with a 96-hour sequence and a deeper stack is the proper deep-learning baseline. Our reading of the recent tabular-forecasting literature (Grinsztajn et al., 2022) is that this would not change the qualitative ordering, but the gap would close.

8 Data and code release

The IMD-4 per-commune scores ($n = 34{,}858$) and per-station scores on the calibration panel ($n = 5{,}442$) are released under the Open Database Licence at https://github.com/rohanfosse/imd-national-catalogue. The predictive pipeline of Section 4 – including the LightGBM, Ridge, ARIMA, persistence and LSTM model definitions, the Vélomagg trip-feature engineering scripts and the benchmark runner – is released under MIT at the same URL. Trip data is derived from the public Vélomagg dock-based GBFS feed and the Vélomagg open-data portal, archived with SHA-256 hashes for bit-exact reproduction. Weather features are from Open-Meteo, archived likewise.

9 Conclusion

We introduced the IMD-4, a Bayesian composite indicator of cycling-environment quality computable on every French commune from public open data, and benchmarked its predictive value against five baselines on a five-year hourly bike-share trip panel. An off-the-shelf LightGBM regressor augmented with the 13 IMD station-level features achieves $R^2_{\text{trips}} = 0.311$ on a one-year temporal hold-out, against 0.042 for the same regressor without the IMD features (+0.27 R^2 gap). The IMD-augmented model matches a station-fixed-effect baseline within 0.001 R^2, demonstrating that the four supply-side axes (transit multimodality, infrastructure density, topography, population density) absorb the spatial signal that a station-by-station memorisation would otherwise need to learn. Classical time-series baselines (persistence, ARIMA) and a deep-learning baseline (LSTM) are materially outperformed.

The IMD-4 is computable on all 34,858 French communes, so the pipeline transfers to any commune with an active dock-based network. The natural follow-up is a multi-city benchmark on the French GBFS portfolio, which we leave for a v1.1 release of the indicator.

Data and code availability

The full pipeline – IMD-4 calibration, national application, Vélomagg trip-feature engineering and the six-model benchmark runner – is released under MIT at `https://github.com/rohanfosse/imd-national-catalogue`. The per-commune IMD-4 scores ($n = 34{,}858$) and per-station scores on the calibration panel ($n = 5{,}442$) are released as versioned Parquet files at the same URL under ODbL. Trained LightGBM and LSTM model artefacts are released alongside the benchmark runner. All raw inputs (Vélomagg GBFS feed, data.gouv *linéaire cyclable* and *stations TC*, Open-Meteo weather extracts, INSEE Filosofi and recensement files) are public and archived with SHA-256 hashes in the same repository.

CRediT authorship contribution statement

Rohan Fossé : Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Gaël Pallares :** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GitHub Copilot and Claude (Anthropic) for code completion and editorial language polishing. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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